

HAJIR MUHAMMED

Control Systems · UAV Guidance, Navigation & Control

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Dear Hiring Manager,

I am writing to express my interest in joining Skydagger's engineering team in the area of control, guidance, and navigation systems. As a final-year Control Engineering student at Yıldız Technical University with hands-on UAV development experience, I am especially motivated by Skydagger's rapid emergence as a serious force in autonomous fixed-wing platforms and the company's ambition to define the next generation of Turkish UAV technology.

What sets Skydagger apart for me is the directness of its engineering. The *TOYCA-05* fixed-wing platform, the *AVCI* interceptor's combination of quadrotor agility with missile-class velocity, and the deep platform integration with Baykar's TB-series, Kalkan, and Kızılelma represent exactly the class of systems where guidance laws, attitude control, and onboard state estimation determine whether a concept becomes a deployable capability. These are the disciplines I have spent my academic career preparing to contribute to.

My senior thesis, *Design and Control of a VTOL UAV*, took me end-to-end through the work an autonomous-flight engineer does in practice. I designed and 3D-printed the airframe, derived a full 6-DOF nonlinear model in MATLAB/Simulink, performed system identification on the propulsion and airframe response, and designed and tuned cascaded attitude and altitude controllers using PID and state-space methods. I implemented a sensor-fusion algorithm to estimate orientation from raw IMU data, and deployed the resulting control logic on an STM32 microcontroller with a SpeedyBee flight controller running iNAV. I am equally comfortable in the Ardupilot and Mission Planner ecosystem.

Beyond the thesis, I have completed rigorous laboratory work in DC motor control, magnetic levitation, ball-and-beam balancing, and coupled-tank level control, and have competed in the *MATLAB Mini Drone Competition* designing trajectory-tracking flight control for a simulated quadrotor. Together with strong coursework in linear and digital control, flight mechanics, signal processing, and embedded systems, this background has given me the analytical foundation — and the practical reflexes — to contribute from day one.

I would welcome the opportunity to discuss how my engineering background can support Skydagger's work in autonomous flight algorithms and the rapid integration cycles your platforms demand. Thank you for your time and consideration.

Respectfully,

Hajir Mohammed